

TRAINING CHESS BOTS WITH VARIOUS MACHINE LEARNING TECHNIQUES

by

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I declare that this dissertation is my own work and that the work of others is acknowledged and indicated by explicit references.

Bhavya Singh
May 2026

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Abstract

Write a summary of the work presented in your dissertation. Introduce the topic and highlight your main contributions and results. The abstract should be comprehensible on its own, and should not contain any references. As far as possible, limit the use of jargon and abbreviations, to make the abstract readable by non-specialists in your area. Do not exceed 300 words.

Acknowledgements

Write any personal words of thanks here. Typically, this space is used to thank your supervisor for their guidance, as well as anyone else who has supported the completion of this dissertation, for example by discussing results and their interpretation or reviewing write ups. It is also usual to acknowledge any financial support received in relation to this work.

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Glossary

A_f	The source message, being a sequence of f source symbols $a_1a_2 \dots a_f$
a_i	The i^{th} symbol in the source message, where $a_i \in S_m$
B_g	The decoded message, being a sequence of g source symbols $b_1b_2 \dots b_g$
b_i	The i^{th} symbol in the decoded message, where $b_i \in S_m$
C_h	The transmitted (compressed) message, being a sequence of h Tunstall codewords $c_1c_2 \dots c_h$
c_i	The i^{th} codeword in the transmitted message, where $c_i \in T_n$
D_h	The received (compressed) message, which for a complete Tunstall code is a sequence of h Tunstall codewords $d_1d_2 \dots d_h$
d_i	The i^{th} codeword in the received message, where $d_i \in T_n$ for a complete Tunstall code

Abbreviations

BER	Bit Error Rate
BPSK	Binary Phase Shift Keying
BSC	Binary Symmetric Channel
DCT	Discrete Cosine Transform
ECC	Error Correcting Codes
FEC	Forward Error Correction
JPEG	Joint Photographic Experts Group
MPEG	Moving Pictures Experts Group
SER	Symbol Error Rate
SNR	Signal to Noise Ratio

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Bibliography

- Briffa, J. A., Schaathun, H. G. & Wesemeyer, S. (2010), An improved decoding algorithm for the Davey-MacKay construction, *in* ‘Proc. IEEE Intern. Conf. on Commun.’, Cape Town, South Africa.
- Burrows, M. & Wheeler, D. J. (1994), A block-sorting lossless data compression algorithm, Technical report, Digital SRC Research Report.
- el Gamal, A. A., Hemachandra, L. A., Shperling, I. & Wei, V. K. (1987), ‘Using simulated annealing to design good codes’, *IEEE Transactions on Information Theory* **33**(1), 116–123.
- Farrell, P. G. (1992), ‘Notes on source coding’. University of Manchester.
- Henderson, B. (2009), *Netpbm*.
URL: <http://netpbm.sourceforge.net/doc/>
- NVI (2009), *NVIDIA CUDA Programming Guide*. Version 2.3.
- Press, W. H., Teukolsky, S. A., Vetterling, W. T. & Flannery, B. P. (1992), *Numerical Recipes in C: The Art of Scientific Computing*, second edn, Cambridge University Press.
- Tunstall, B. P. (1968), Synthesis of Noiseless Compression Codes, PhD thesis, Georgia Institute of Technology.